

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919425, or equivalent

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

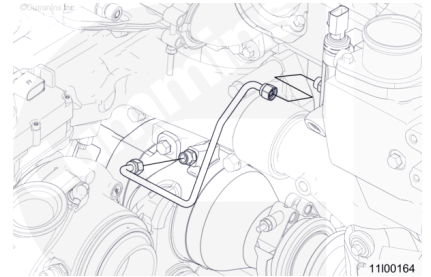
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html)
- Remove the air intake piping. See equipment manufacturer service information. Use protective caps from the Air Handling Clean Care Kit, Part Number 4919425, to cover the connection points.

- Remove the selective catalytic reduction (SCR) aftertreatment coolant supply hose and fittings from the exhaust gas recirculation (EGR) cooler coolant return tube. See equipment manufacturer service information.

Remove

⚠ CAUTION ⚠

Be sure all the components are clean and free of debris that can cause damage to o-ring seals.

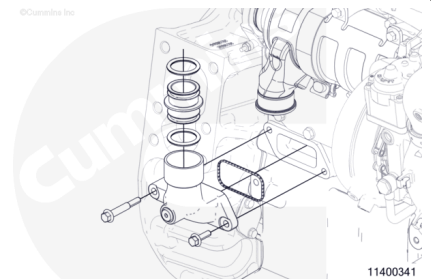


Note : Use a second wrench to hold the male unions on the thermostat housing and on the EGR cooler while removing the EGR cooler coolant vent line fittings.

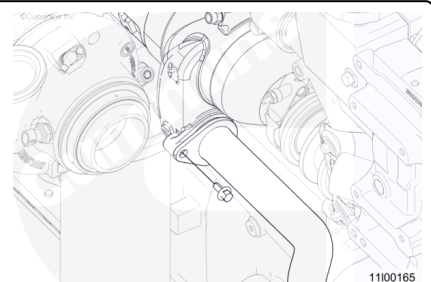
Disconnect the EGR cooler coolant vent line from the thermostat housing and the EGR cooler.

Remove EGR cooler coolant return tube mounting capscrews.

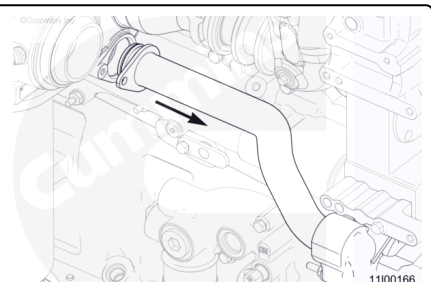
Remove the EGR cooler coolant return jumper tube from the EGR cooler.



Remove the EGR cooler coolant supply tube mounting capscrew.



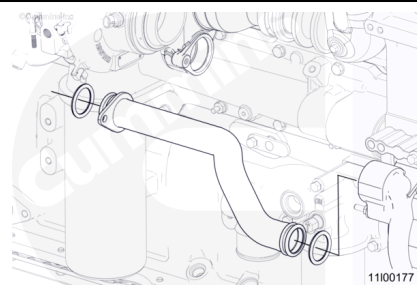
Slide the EGR cooler coolant supply tube toward the water pump housing until the coolant supply tube is clear of the EGR cooler coolant inlet.



Rotate the EGR cooler coolant supply tube away from the engine.

Remove the EGR cooler coolant supply tube from the engine.

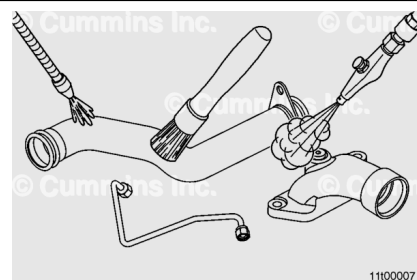
Remove and discard the o-rings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

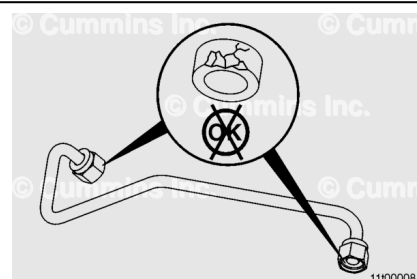
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the EGR cooler coolant vent line and the EGR cooler coolant supply and return tubes with safety solvent.

Dry with compressed air.

Inspect the EGR cooler vent line grommets for damage.

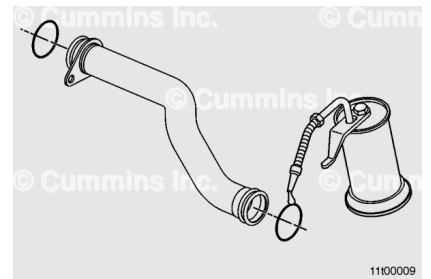
Replace the grommets if damage is found.



Install

⚠ CAUTION ⚠

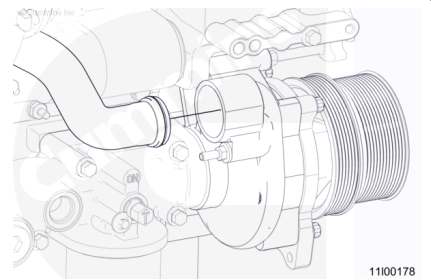
Do not use petroleum products on the EGR cooler coolant line o-rings. The o-rings will swell and cause the connection to leak.



Install new o-rings on the EGR cooler coolant supply tube.

Lubricate the o-rings with clean engine coolant, soapy water, or vegetable oil, to aid in installation.

Insert the EGR cooler coolant supply tube into the water pump outlet until the EGR cooler coolant supply tube is clear of the EGR cooler coolant inlet.

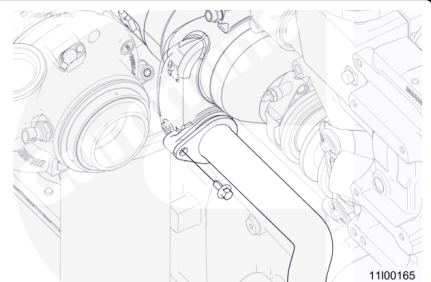


Rotate the EGR cooler coolant supply tube toward the engine until it lines up with the EGR cooler coolant inlet.

Insert the EGR cooler coolant supply tube into the EGR cooler coolant inlet.

Install and tighten the EGR cooler coolant supply capscrew.

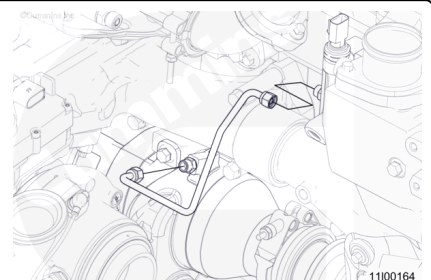
Torque Value: 20 n•m [177 in-lb]



Note : Use a second wrench to hold the male unions on the thermostat housing and on the EGR cooler while installing the EGR cooler coolant vent line fittings.

Install the EGR cooler coolant vent line to the thermostat housing and the EGR cooler.

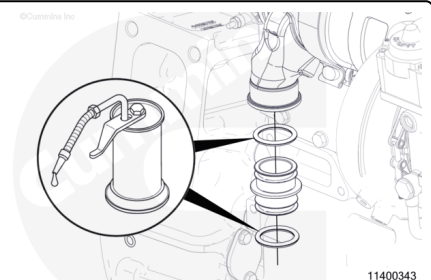
Torque Value: 14 n•m [124 in-lb]



Install the o-rings on the coolant return tube jumper tube.

Lubricate the o-rings with clean engine coolant, soapy water, or vegetable oil to aid in installation.

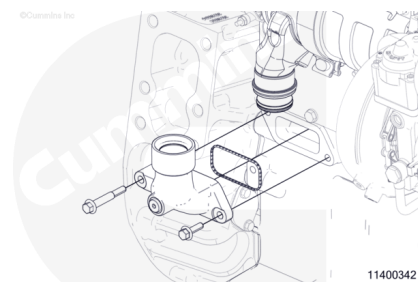
Install the jumper tube into the EGR cooler.



Install the press-in-place seal and the coolant return tube.

Install and tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Remove the protective caps from the intake piping.
- Install the intake piping. See equipment manufacturer service information.
- Install the SCR aftertreatment coolant supply hose and fittings to the EGR cooler coolant return tube. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks and proper operation.